BIOCHEMISTRY CUME
APRIL 4, 2003

On March 10, 2003 Dr. David Weber from the University of Maryland gave a seminar. Answer the following questions about the seminar.

1. Dr. Weber described work on S100 proteins, an important and high conserved family of regulatory proteins. Dr. Weber spent considerable time discussing the interaction of these S100 proteins with another well studied protein, p53. What is the importance of this second protein or what is the overall significance of the research program? One sentence answer. 15 points

2. Dr. Weber described in great detail the conformational changes that S100 undergoes which a small molecule signal binds to it. What small molecule was this? 10 points

3. Describe in general terms the conformational change that S100 undergoes upon binding this small molecule and how this alters interaction with p53. 30 points

4. What experimental technique did Dr. Weber's group principally use to determine these structural changes? 15 points

4. Dr. Weber then described rational drug design efforts. Describe very briefly and in general terms how the drug was designed. 15 points.

5. Again, very briefly and in general terms, describe the proposed mechanism by which the drug was supposed to alter the activity of S100. 15 points